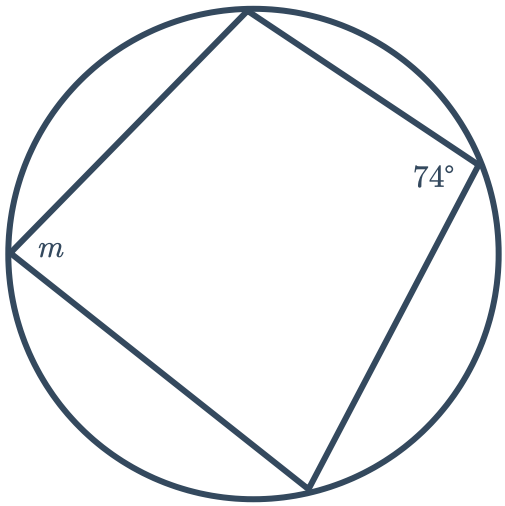
Cyclic quadrilaterals

1 The diagram shows a cyclic quadrilateral. Solve for m.

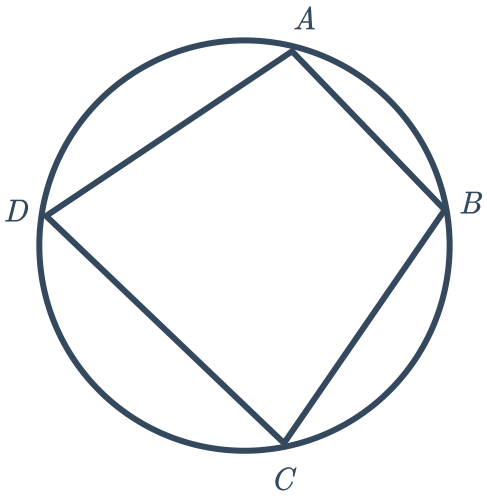


2 In the following figure, $ABCD$ is inscribed inside a circle:

a What is the sum of any two opposite angles in the quadrilateral $ABCD$?

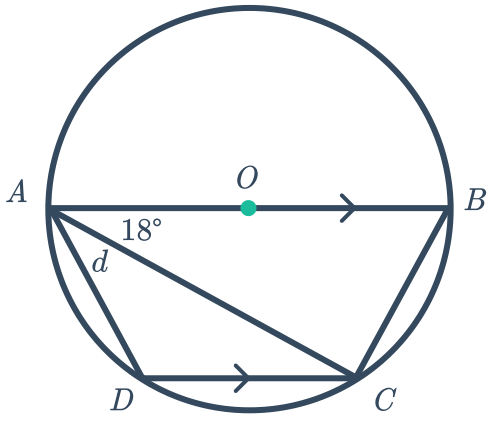
b Can a parallelogram be inscribed inside a circle? Explain your answer.

c Can a rectangle be inscribed inside a circle? Explain your answer.



3 Consider the following figure:

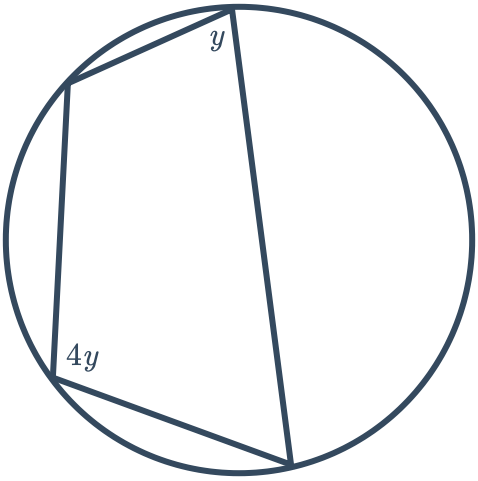
Determine the value of d.



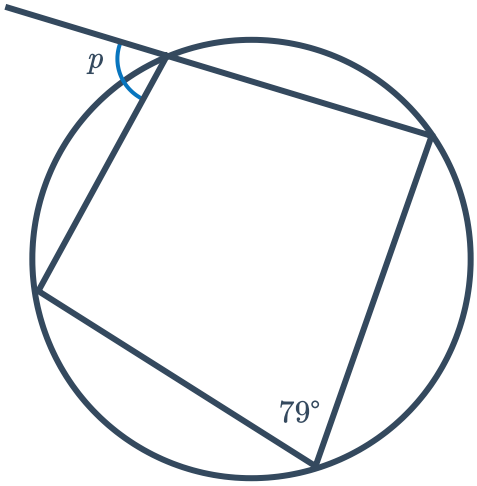
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4 Find the value of the pronumerals in the following diagrams. Give reasons for your answer.

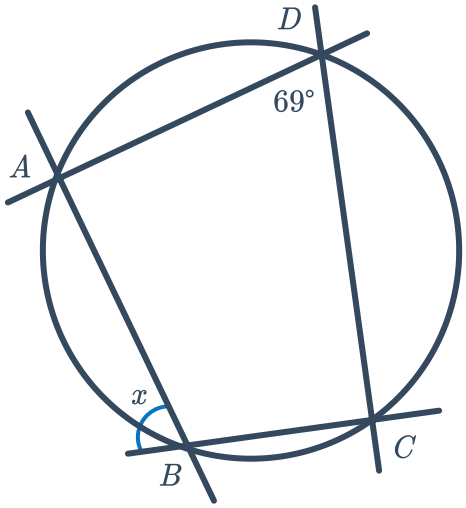
a



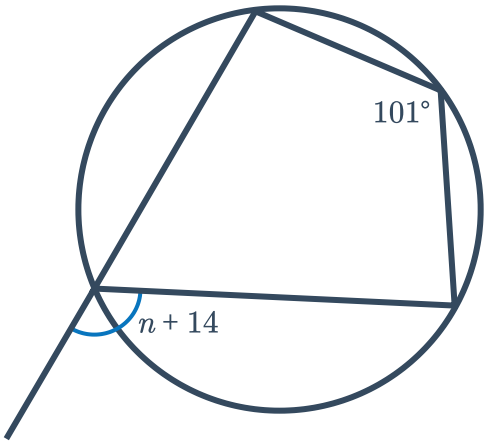
b



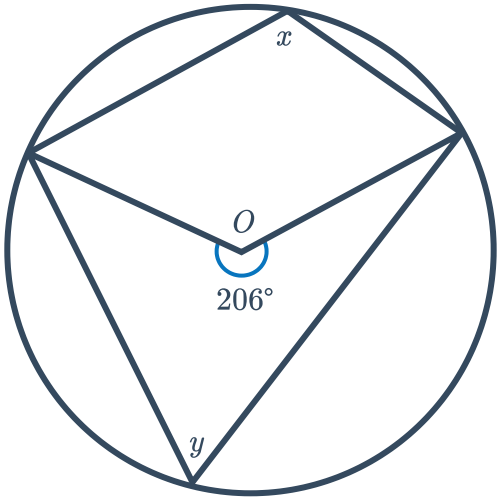
c



d

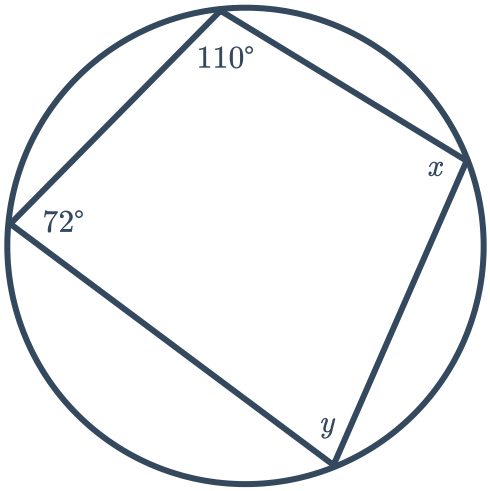


5 Find the value of y in the following diagram:



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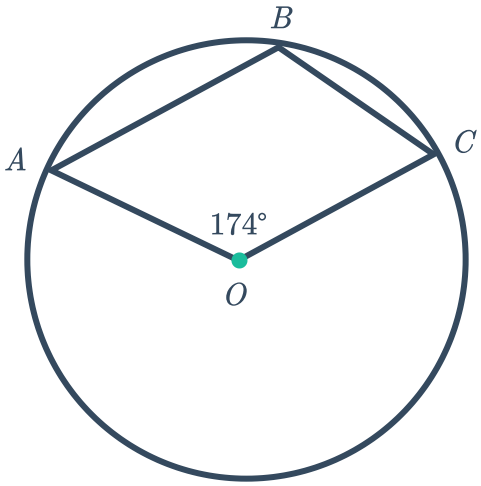
6 Consider the following figure:



a Find the value of x.

b Find the value of y.

7 In the diagram, $\angle AOC = 174^\circ$, where O is the centre of the circle:

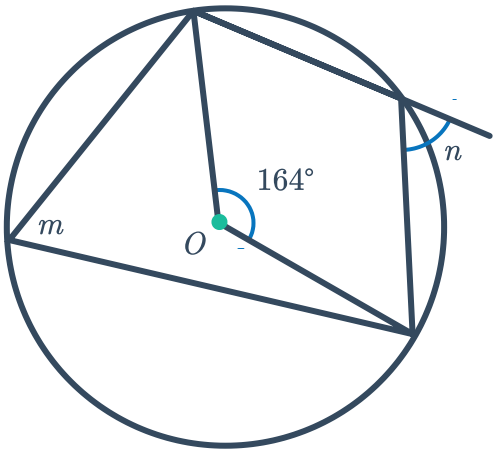


a Determine the size of reflex $\angle AOC$.

b Determine the size of $\angle ABC$.

c Can a circle be drawn through the vertices of quadrilateral $OABC$? Explain your answer.

8 In the diagram, O is the centre of the circle:

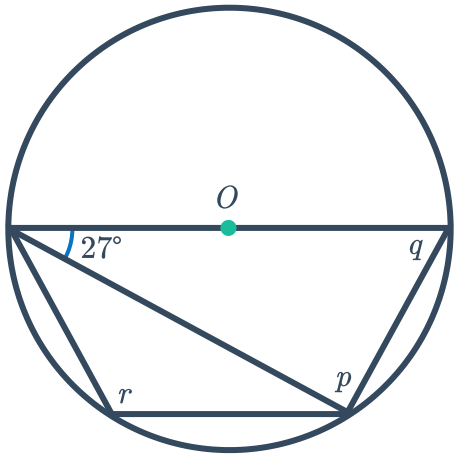


a Solve for the value of m, giving reasons for your answer.

b Solve for the value of n, giving reasons for your answer.

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9 In the diagram, O is the centre of the circle.

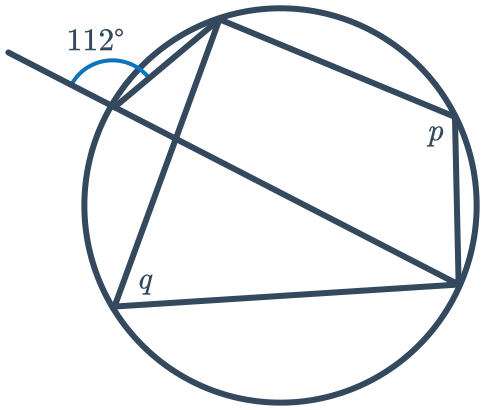


a Solve for p, giving reasons for your answer.

b Solve for q, giving reasons for your answer.

c Solve for r, giving reasons for your answer.

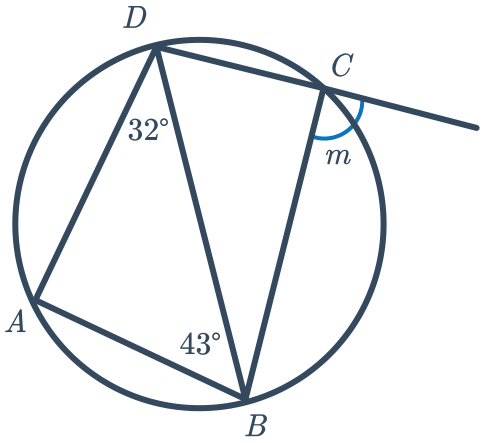
10 Consider the given diagram:



a Find the value of p, giving reasons for your answer.

b Find the value of q, giving reasons for your answer.

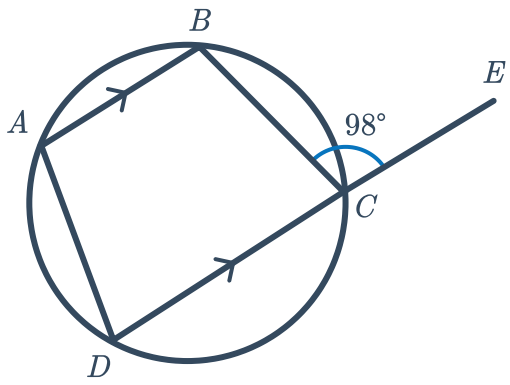
11 In the diagram, $\angle ADB = 32^\circ$ and $\angle DBA = 43^\circ$. Find the size of the angle marked m, giving reasons for your answer.



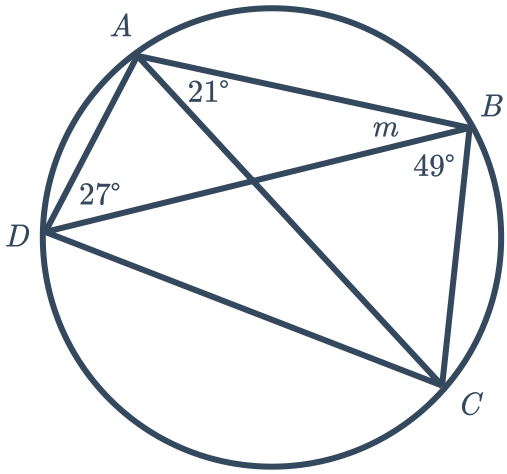
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12 In the diagram, $AB \parallel DC$ and $\angle BCE = 98^\circ$. DC is produced to point E.

Find the size of $\angle BAD$.

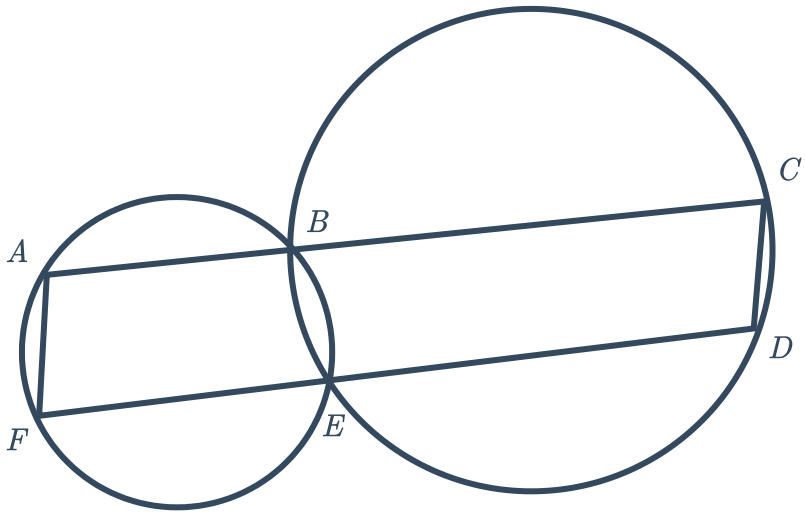


13 In the diagram, $\angle DBA = 27$, $\angle CAB = 49$, $\angle CBD = 21$ and $\angle ABD = m$. Solve for m, giving reasons.



14 In the following figure, consider the two circles intersect at points B and E.

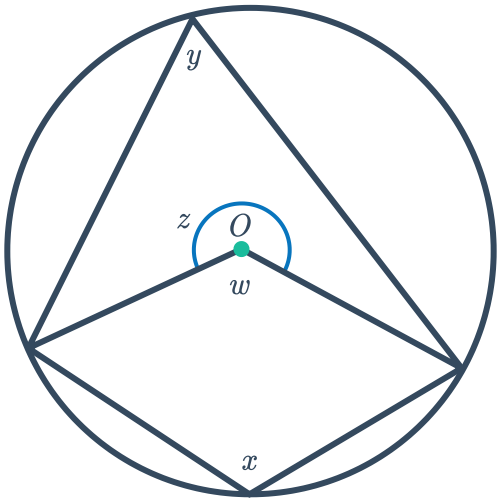
If $\angle BCD = 94^\circ$. Find $\angle BAF$.



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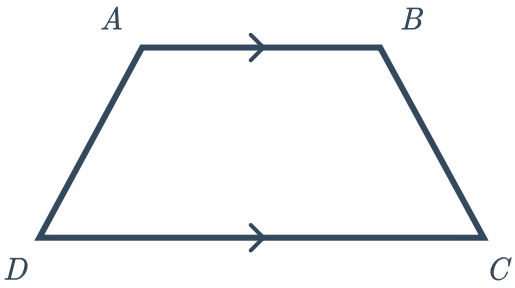
Proofs

15 In the diagram, O is the centre of the circle. Show that x and y are supplementary angles.



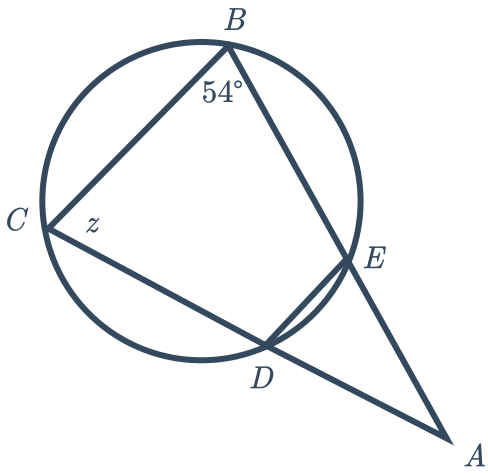
16 In the diagram, $ABCD$ is an isosceles trapezium, so $AD = BC$.

Prove that the points A, B, C and D are concyclic.



17 Consider the following diagram:

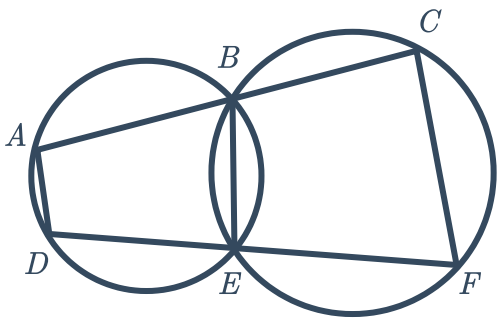
Prove that $\angle BAC = 126 - z$.



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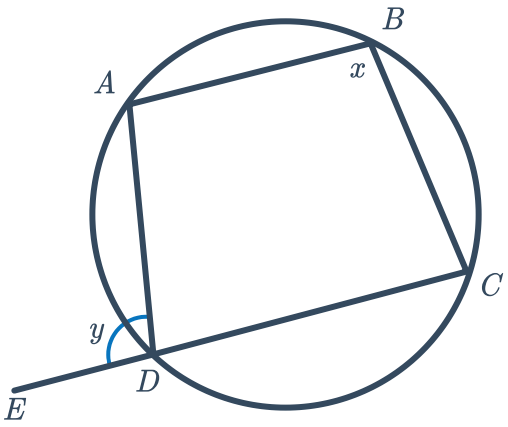
18 Consider the following diagram:

Prove $AD||CF$.



19 Consider the following diagram:

Prove that $x = y$.

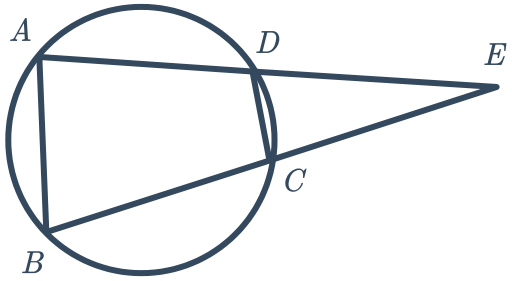


20 Consider the figure:

a Prove that $\angle ABC = \angle CDE$.

b By proving two similar triangles, Prove that $\angle BAD$ and $\angle DCE$ are equal.

c Hence prove that $EB \times EC = ED \times EA$.



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